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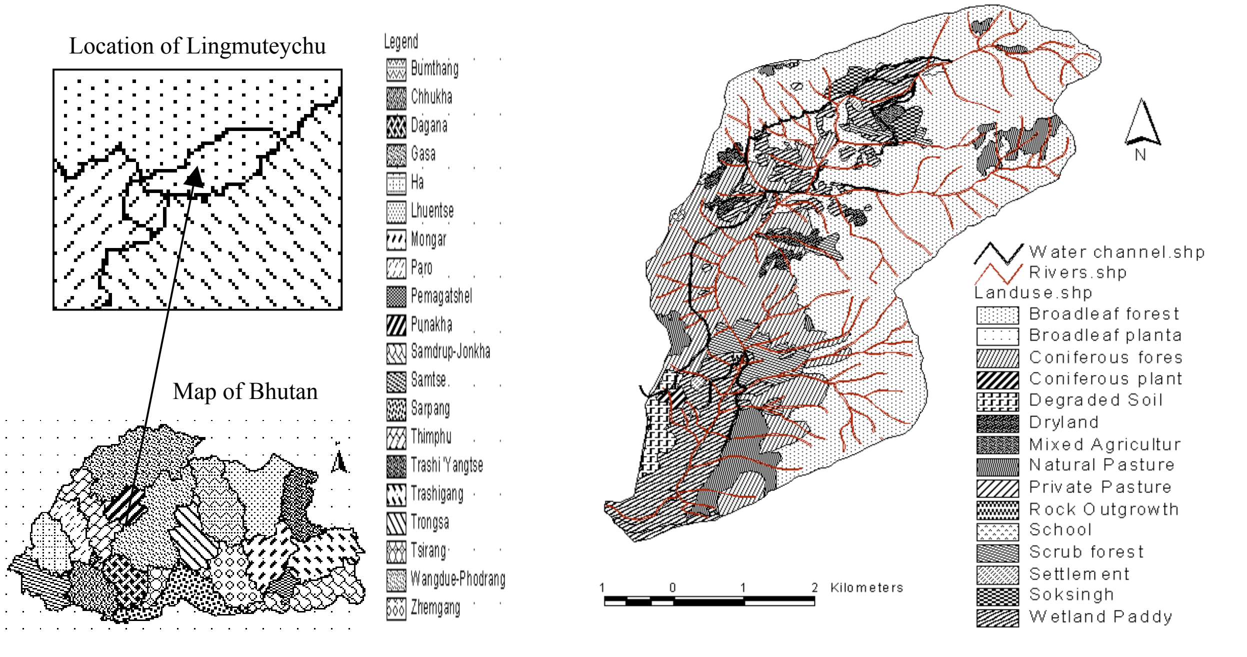


Figure 1. Map of Study Site - Lingmuteychu watershed. (Source: LUSS, 2003)

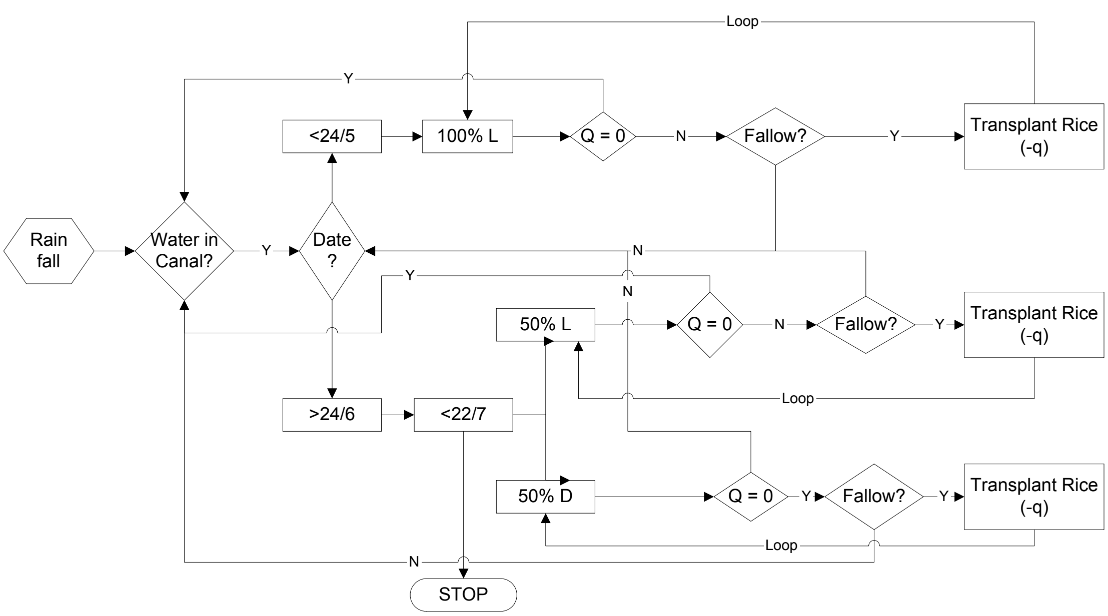


Figure 2. Activity Diagram during transplanting in Linbukha and Dompola

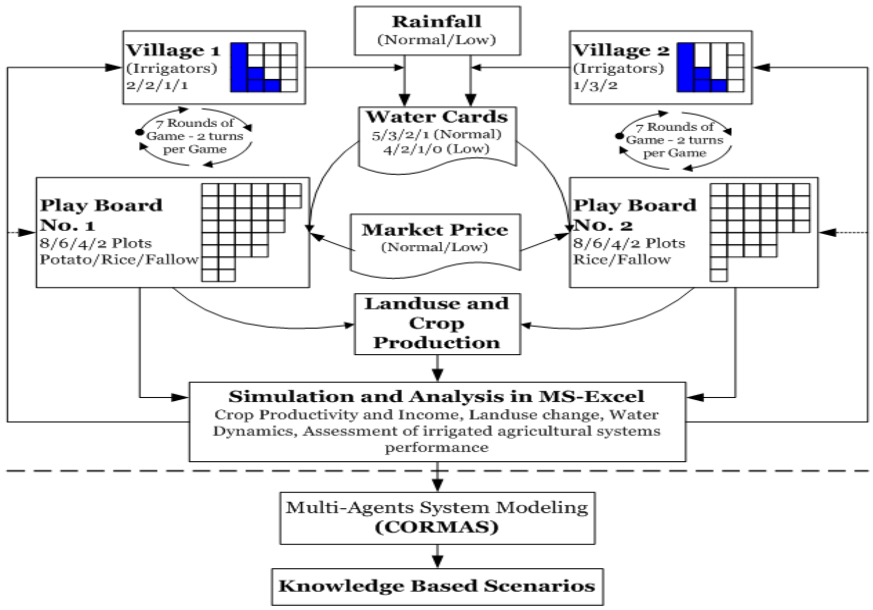


Figure 3. Conceptual Framework for RPG Process

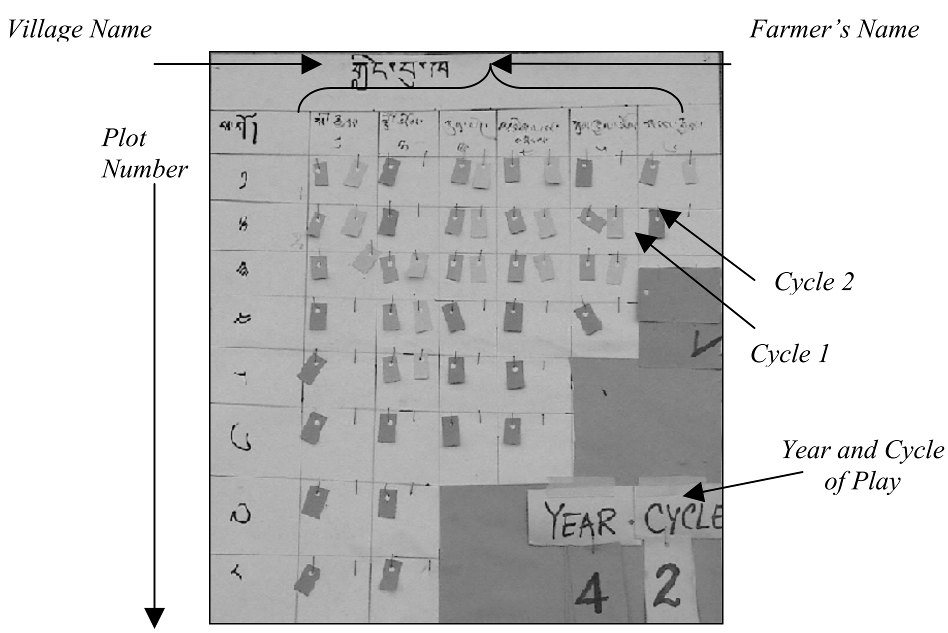
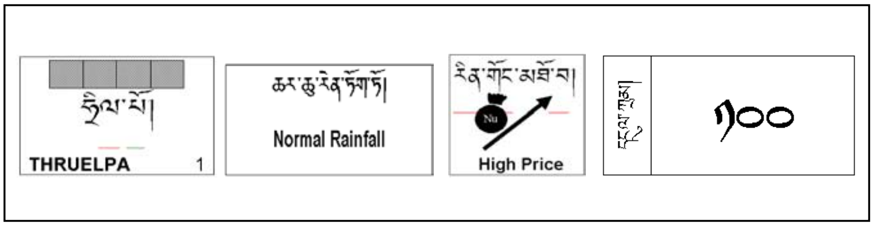
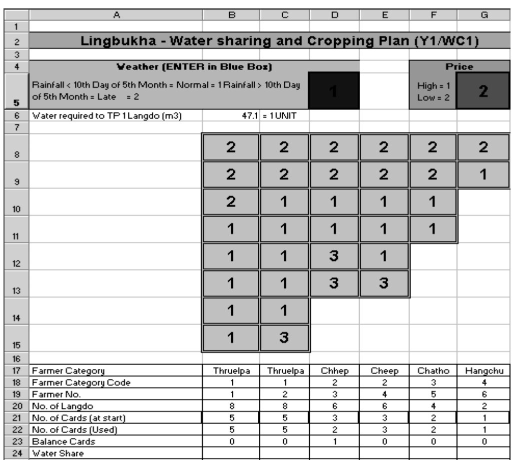


Figure 4. RPG Playing Board

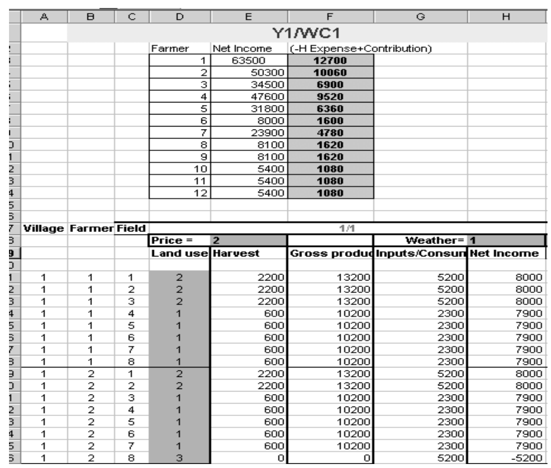


(a) (b) (c) (d)

Figure 5. Cards used in the game : (a). Farmer Category; (b). Rainfall (Chance card); (c). Market Price (Chance card); (d). Cash



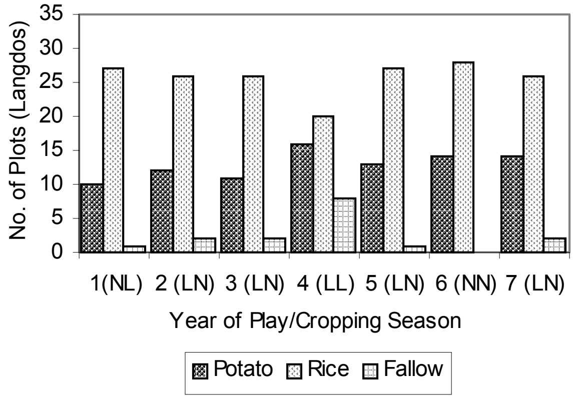
	A	B	C	D	E	F	G
1							
2	Lingbukha - Water sharing and Cropping Plan (Y1/WC1)						
3							
4	Weather (ENTER in Blue Box)					Price	
5	Rainfall < 10th Day of 5th Month = Normal = 1 Rainfall > 10th Day of 5th Month = Late = 2			1		High = 1 Low = 2	2
6	Water required to TP 1 Langdo (m3)	47.1	= 1 UNIT				
7							
8		2	2	2	2	2	2
9		2	2	2	2	2	1
10		2	1	1	1	1	
11		1	1	1	1	1	
12		1	1	3	1		
13		1	1	3	3		
14		1	1				
15		1	3				
16							
17	Farmer Category	Thruelpa	Thruelpa	Chhep	Cheep	Chatho	Hangchu
18	Farmer Category Code	1	1	2	2	3	4
19	Farmer No.	1	2	3	4	5	6
20	No. of Langdo	8	8	6	6	4	2
21	No. of Cards (at start)	5	5	3	3	2	1
22	No. of Cards (Used)	5	5	2	3	2	1
23	Balance Cards	0	0	1	0	0	0
24	Water Share						



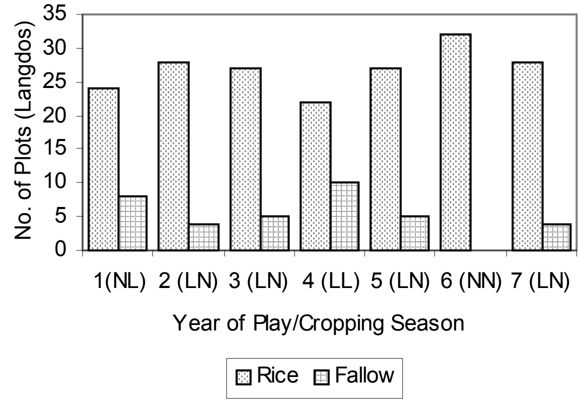
A	B	C	D	E	F	G	H
	Y1/WC1						
			Farmer	Net Income	(-H Expense+Contribution)		
			1	63500	12700		
			2	50300	10060		
			3	34500	6900		
			4	47600	9520		
			5	31800	6360		
			6	8000	1600		
			7	23900	4780		
			8	8100	1620		
			9	8100	1620		
			10	5400	1080		
			11	5400	1080		
			12	5400	1080		
Village	Farmer	Field			1/1		
			Price =	2		Weather=	1
			Land use	Harvest	Gross produc	Inputs/Consun	Net Income
1	1	1	2	2200	13200	5200	8000
1	1	2	2	2200	13200	5200	8000
1	1	3	2	2200	13200	5200	8000
1	1	4	1	600	10200	2300	7900
1	1	5	1	600	10200	2300	7900
1	1	6	1	600	10200	2300	7900
1	1	7	1	600	10200	2300	7900
1	1	8	1	600	10200	2300	7900
1	2	1	2	2200	13200	5200	8000
1	2	2	2	2200	13200	5200	8000
1	2	3	1	600	10200	2300	7900
1	2	4	1	600	10200	2300	7900
1	2	5	1	600	10200	2300	7900
1	2	6	1	600	10200	2300	7900
1	2	7	1	600	10200	2300	7900
1	2	8	3	0	0	5200	-5200

(a) (b)

Figure 6. MS Excel Spreadsheet used in the RPG (a) Play-board (b) Simulation



(a)



(b)

1-7 in X-Axis indicate Year; Letters in parenthesis indicates rainfall pattern in two cycles, i.e. N: Normal rainfall and L: Low rainfall (e.g. NL means first cycle rainfall was "Normal" and it was "Low" in second cycle).

Figure 7. Land-use changes in (a) Linbukha and (b) Dompola over the cropping seasons during the first gaming session.

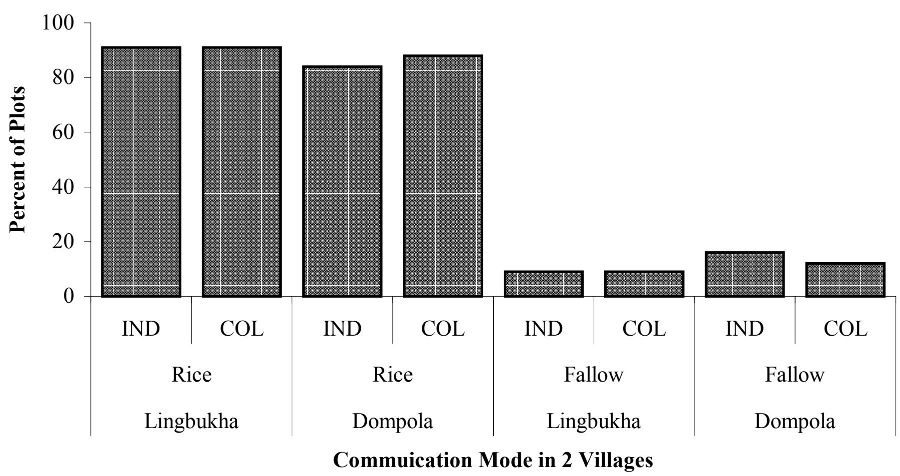
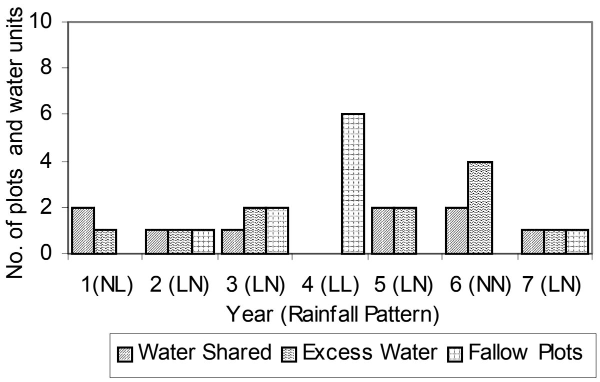
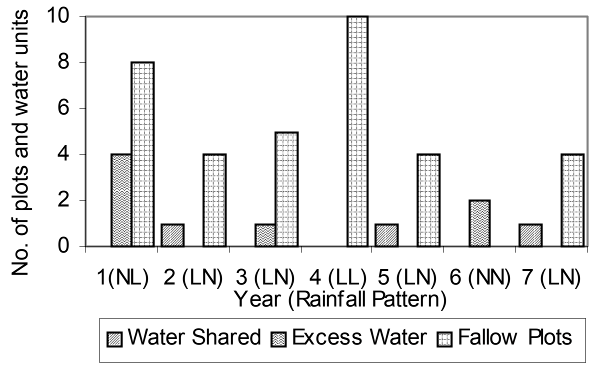


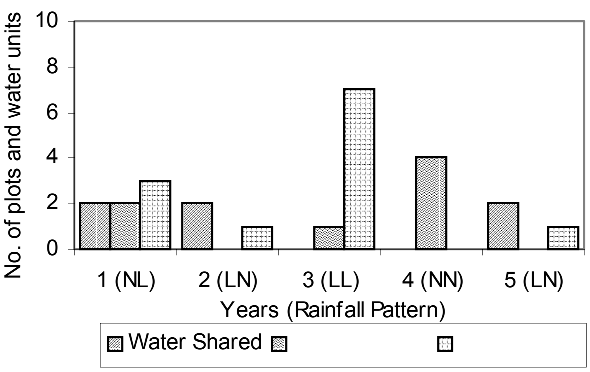
Figure 8. Comparison of land-use changes in two villages due to different modes of communication. (IND: Intra-village mode of communication; COL: Intra-village mode of communication)



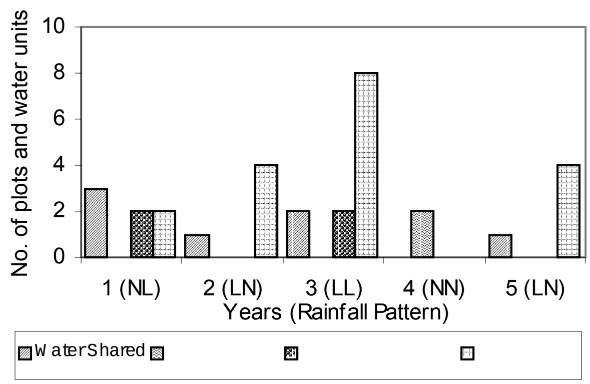
(a) Linbukha-Intra Village Communication mode



(b) Dompola-Intra Village Communication mode



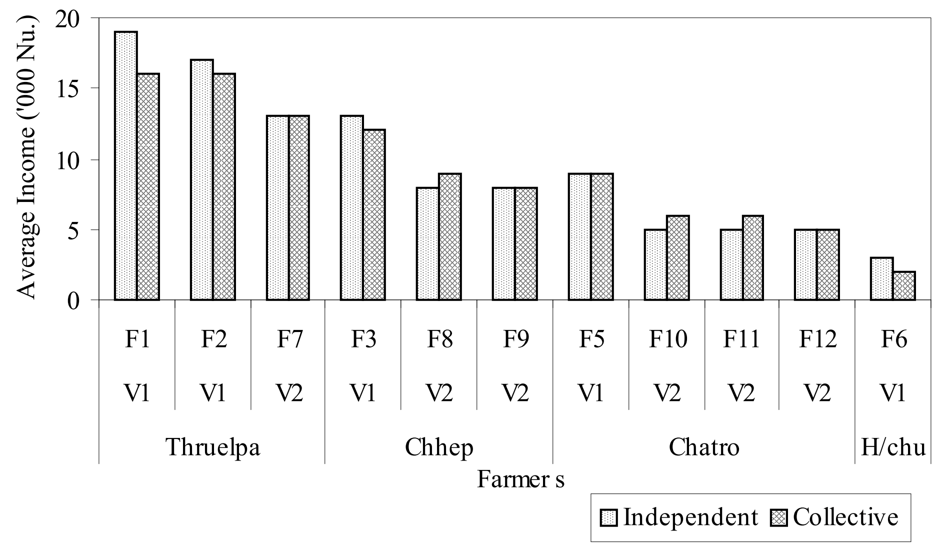
(c) Linbukha-Inter Village Communication mode



(d) Dompola-Inter Village Communication mode

1-7 in X-Axis indicate Year; Letters in parenthesis indicates rainfall pattern in two cycles, i.e. N: Normal rainfall and L: Low rainfall (e.g. NL means first cycle rainfall was "Normal" and "Low" in second cycle); From Other Village in Figure (d) means water shared between village.

Figure 9. Water Dynamics in two villages following two modes of communications



F1-F12 imply farmers coded as 1-12; V1=Linbukha; V2=Dompola; and Thruelpa, Chhep, Chatro and Hangchu indicate farmer category based on water share.

Figure 10 : Income variation among farmer categories according to two modes of communication.

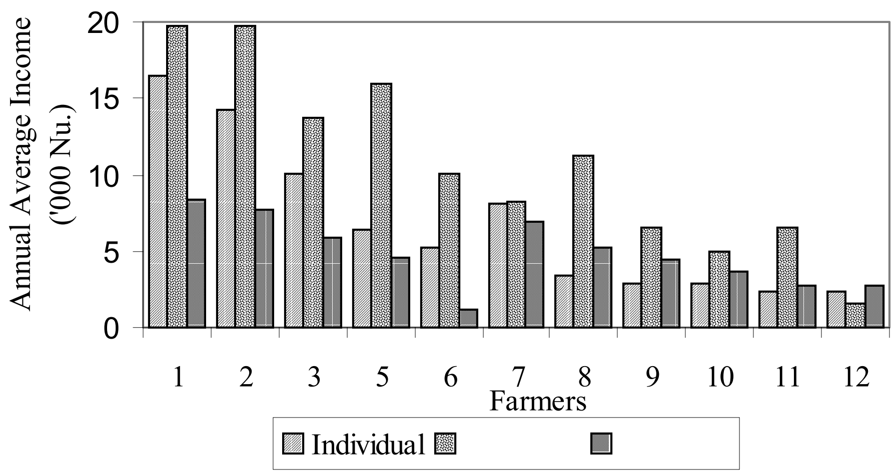


Figure 11. Annual income generated from three different modes of communication

Table 1. Farmers' response on swapping the role between two villages

Responses	Percent Respondent (n=11)
Did not like swapping the role (Farmer 2)	9%
Comparison of income between two villages (1, 3, 8)	27%
Understood the potential of potato in Linbukha and are motivated to try out in Dompola (9 & 11)	18%
Resource advantage of Linbukha farmers (10)	9%
Understood the problem of Dompola farmers (12)	9%
Possibility to grow potato in Linbukha (5, 6, 7)	27%